

ACE Workshop: Rare disease genetics and burden testing

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Genetic variation and disease

- Every human genome is approximately 6 billion nucleotides long
- Human genomes are 99.6% identical to each other, however:
 - Each person carries approximately 5 million SNVs as well as other kinds of variation (insertions and deletions, tandem repeats, structural variation)
- The vast majority of this variation is phenotypically and clinically neutral – it does nothing.
- Another large fraction is common and has very small effect size on any human health traits
- Certain examples of rare variation can be catastrophic and cause rare genetic diseases and syndromes

A fundamental component of disease-focused research is distinguishing variants that matter from variants that are unrelated to the trait of interest



Rare Genetic Disease

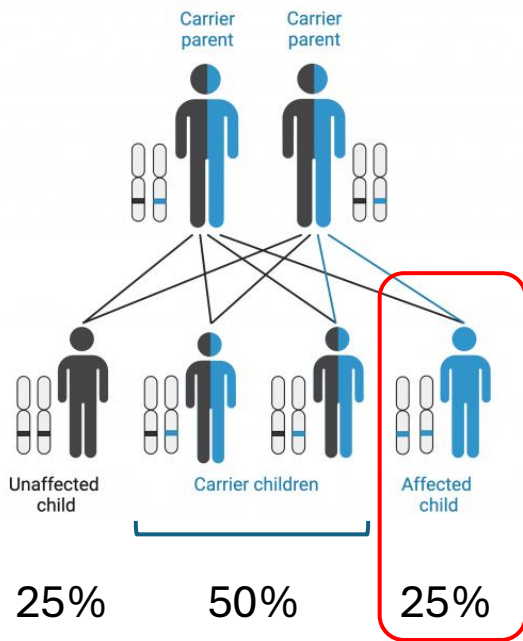
1. Is a given **gene** causally associated with a disease?
2. Is a given **variant** affecting that gene associated with disease?



Definitions:

- **Gene:** the basic functional unit encoded in DNA and expressed into a discrete molecular product (mRNA, protein)
- **Allele/variant:** one of the two or more possible versions of a DNA sequence at a given genomic location (e.g. chr1:55039995-C). Can sometime describe the position itself with or without details of variation known to exist in the human population (e.g. chr1:55039995-C/T, rs334)
- **Diploid:** the presence of two complete sets of chromosomes in an organism's cells, one set inherited from each parental gamete. Humans are diploid.
- **Genotype:** the diploid pair of alleles a given individual carries at a particular genomic location (e.g. AA, AG, GG, CT, ...). Genotypes may be homozygous (identical to each other) or heterozygous (different from each other) in a person.
- **Reference genome:** an accepted representation of the human genome sequence (version controlled; GRCh38 is common) that is used by researchers as a standard for comparison to DNA sequences generated in their studies. The reference genome is not a “perfect and healthy” ideal human and because it is concretely ACTG at all loci, it contains alleles that may be associated with increased and/or decreased risk for common diseases, as well as versions of genes that may predict certain traits. The reference genome is not a real person.

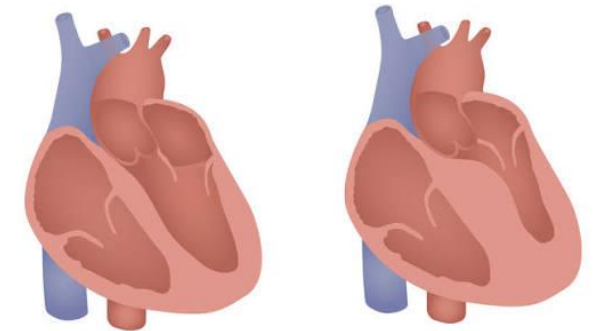
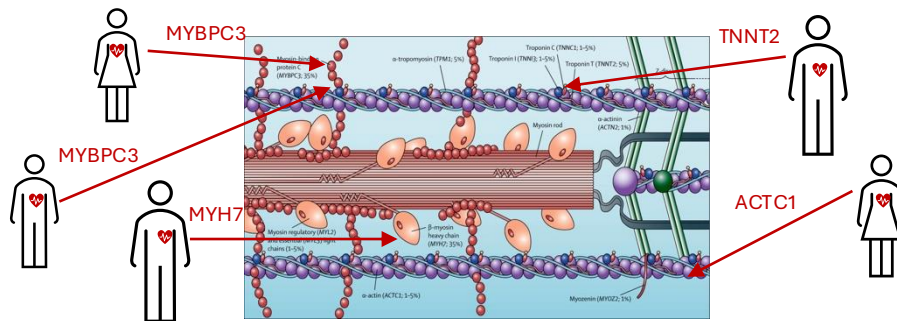
Principles of rare genetic disease association



1. The causal variant must be rare
2. The causal variant must be more common in affected cases than it is in unaffected controls
 - If there is family genetic data available, the causal variant should segregate with the trait in that family
3. The variant must have a functional effect on biology different from the reference
 - Bioinformatics predictions, experimental assays
4. A second, independent line of evidence also supports the gene-disease relationship
 - Other families or studies, animal models

Principles of rare genetic disease association

- **Genetic heterogeneity:** when mutations in some number of different genes can all cause the same clinical diagnosis

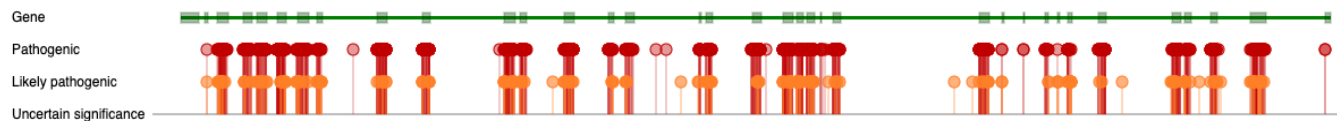


Normal

Hypertrophic Cardiomyopathy

- **Allelic heterogeneity:** when people with the clinical diagnosis and same causal gene have different pathogenic alleles

MYBPC3:



Conceptual foundation
for burden testing

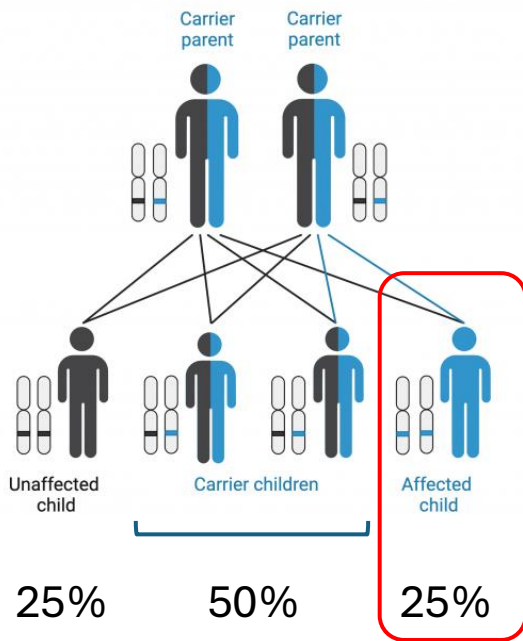
Burden Test and Sequence Kernel Association (SKAT) Tests for Gene-Disease Association

- Assumes allelic heterogeneity, and uses the gene – rather than the variant – as the unit for association
- **Burden test:** collapses all qualifying variants to the gene level prior to association testing
- **SKAT:** uses a variance component statistic (weighted by inverse frequency) that creates a patient-to-patient similarity score for the gene. Aka, if two patients both carry an ultra-rare variant within a gene, they are mathematically rated as two *highly similar* individuals.
- **SKAT-o (SKAT-optimal):** combines burden + SKAT

Statistical model comparison

Feature	Burden Test (more intuitive)	SKAT tests (more sophisticated)
Core Mechanism	Collapses/aggregates variants into a single genetic score <i>before</i> testing association	Aggregates individual variant test statistics using pair-wise genotypic similarity
Effect Direction Assumption		
Non-causal variants		
Optimal scenario		
Statistical framework		
Non-centrality parameter		

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Variant annotation: how to know if a variant is rare

- Start with your disease/trait, not genetics
- If the disease is well studied, review known cohort studies and previously reported pathogenic variants as a guide
 - ClinVar: <https://www.ncbi.nlm.nih.gov/clinvar/>
 - Your own cohort
- Leverage large public resources
 - gnomAD (see also GenIE): <https://gnomad.broadinstitute.org/>
(<https://gnomad.broadinstitute.org/news/2024-06-genie/>)
 - All of Us: <https://databrowser.researchallofus.org/snvsindels/>
- Pay attention to ancestry
 - many variants can be rare in one population, but more frequently observed in another
- Frequency is the most important criteria for ‘qualifying’ a variant

Variant annotation: how to know if a variant is rare

- Hardy-Weinberg equilibrium where:
 - p = major allele frequency
 - q = minor allele frequency

$$1 = p^2 + 2pq + q^2$$

Assume minor allele frequency = 0.01, the genotype frequencies become:

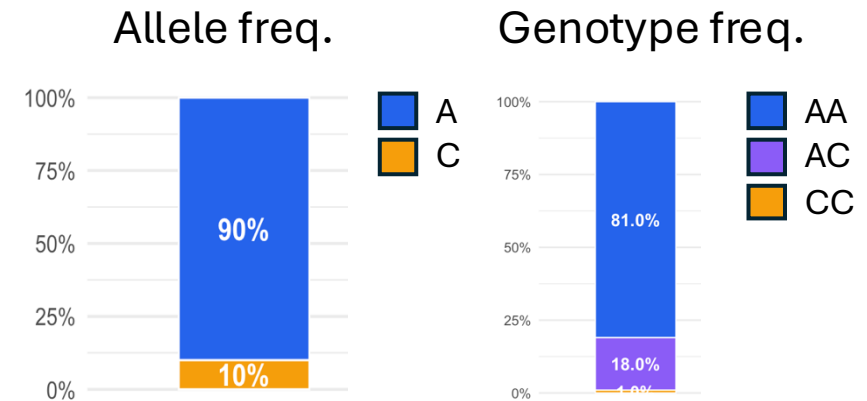
$$1 = (1 - 0.01)^2 + 2(1 - 0.01)(0.01) + (0.01)^2$$

98.01%

1.98%

0.01%

1 in 10,000 people
will be homozygous



ClinVar: <https://www.ncbi.nlm.nih.gov/clinvar/>

- A submitter-supplied database of rare variants interpreted by clinical geneticists for their potential to cause disease.
 - Professional, accredited clinical genetics testing companies
 - Researchers in the field
 - Intended as a way to share information about variants with the broader field
- Observational, with variants observed in real people who have had clinical genetic testing or participated in clinical genetics research studies
- Categorized into ACMG interpretation categories:
 - **Pathogenic**: causes a genetic disease
 - **Likely Pathogenic**: believed to cause disease, but some convincing evidence is missing
 - **Variant of Uncertain Significance**: meets some criteria but no decision could be reached
 - **Likely Benign**: thought not to cause disease, but some convincing evidence is missing
 - **Benign**: does not cause a genetic disease

ClinVar resource: variant page

NM_007294.4(BRCA1):c.5191G>T (p.Glu1731Ter)

Classification
★ ★ ★ ★ ★
Pathogenic (4), Likely pathogenic (1)
1 out of 5 submissions contributed to this classification

No data submitted for somatic clinical impact

No data submitted for oncogenicity

Functional data are available for this variant

Variant Details

Identifiers

NM_007294.4(BRCA1):c.5191G>T (p.Glu1731Ter)
Variation ID: 37646 Accession: VCV00037646.16
Type and length: single nucleotide variant, 1 bp
Location: cytogenetic: 17q21.31 17:43063335 (GRCh38) [NCBI UCSC] 17:41215932 (GRCh37) [NCBI UCSC]

Timeline in ClinVar

First in ClinVar	Last submission	Last evaluated
Germline	Jan 1, 2018	Feb 15, 2025

HGVs

Nucleotide	Protein	Molecular consequence
NM_007294.4:c.5191G>T	NP_055225.1:p.Glu1731Ter	nononsense
NM_001401797.1:c.4579G>T	NP_001394555.1:p.Glu1440Ter	nononsense
NM_001401781.1:c.5257G>T	NP_001394519.1:p.Glu1737Ter	nononsense

Protein change

E1731* E527* E1732* E1684* E1439* E1618* E1642* E1680* E539* E548* E584* E589* E581* E539* E584* E524* E530* E589* E548* E1620* E1603* E1619* E1641* E1649* E1649* E1689* E1689* E1720* E1709* E1730* E1723* E451* E500* E501* E519* E581* E583* E584* E587* E587* E582* E1604* E1620* E1642* E1681* E1728* E1728* E1729* E1730* E514* E519* E539* E549* E550* E580* E581* E582* E582* E520* E520* E1707* E1540* E1640* E1649* E1689* E1689* E1711* E1712* E1720* E1751* E1751* E1753* E400* E519* E539* E549* E560* E560* E560* E560* E563*

Other names

Canonical SPD: NC_000017.11:43063334.C.A
Global minor allele frequency (GMAF):
Allele frequency:
Links:
ClinGen: CA10591159
dbSNP: rs397507244
VefScore

Genes

Gene	OMIM	ClinGen Gene Dosage Sensitivity Curation		Variation Viewer	Related variants	
		HS score	TS score		Within gene	All
BRCA1		Sufficient evidence for dosage pathogenicity	No evidence available	GRCh38 (GRCh37)	14044	16062

Conditions - Germline

Condition	Classification (if of submissions)	Review status	Last evaluated	Variation/condition record
Hereditary cancer-predisposing syndrome	Pathogenic (2)	★ ★ ★ ★	Jun 24, 2025	RCV005711448.5
Hereditary breast ovarian cancer syndrome	Pathogenic (1)	★ ★ ★ ★	Jun 23, 2023	RCV00698353.8
Breast-ovarian cancer; familial, susceptibility to, 1	Pathogenic/Likely pathogenic (2)	★ ★ ★ ★	Nov 13, 2023	RCV001076151.7

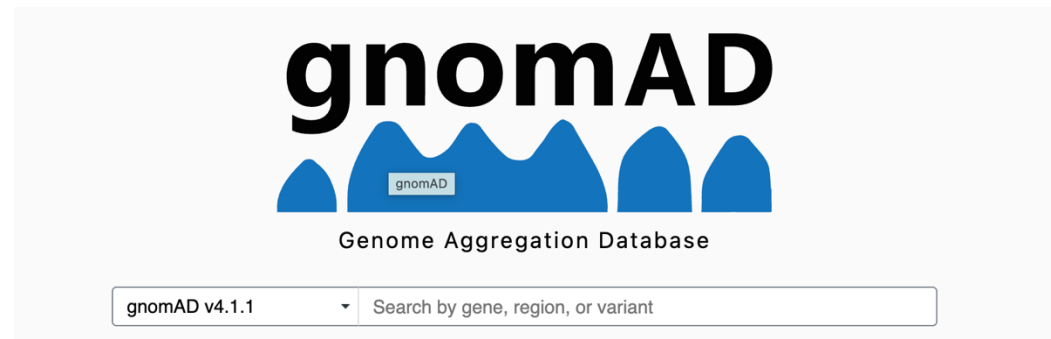
Submissions - Germline

Classification (Last evaluated)	Review status (Assertion criteria)	Condition	Submitter	Collapse all rows
Likely pathogenic (Aug 05, 2023) Contributing to aggregate classification	★ ★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer; familial, susceptibility to, 1	Baylor Genetics Accession: SCV004215036.1 First in ClinVar: Dec 30, 2023 Last updated: Dec 30, 2023	Observation: 1 Collection method: clinical testing Allele origin: unknown Affected status: unknown
Pathogenic (Nov 13, 2023) Contributing to aggregate classification	★ ★ ★ ★ ★ (ACMG Guidelines, 2015)	Breast-ovarian cancer; familial, susceptibility to, 1 (Autosomal dominant inheritance)	All of Us Research Program, National Institutes of Health Accession: SCV004839571.1 First in ClinVar: Apr 20, 2024 Last updated: Apr 20, 2024 Comment: This study involves interpretation of variants in research participants for the purpose of population health screening. Participant phenotype was not available at the time of ... (more)	

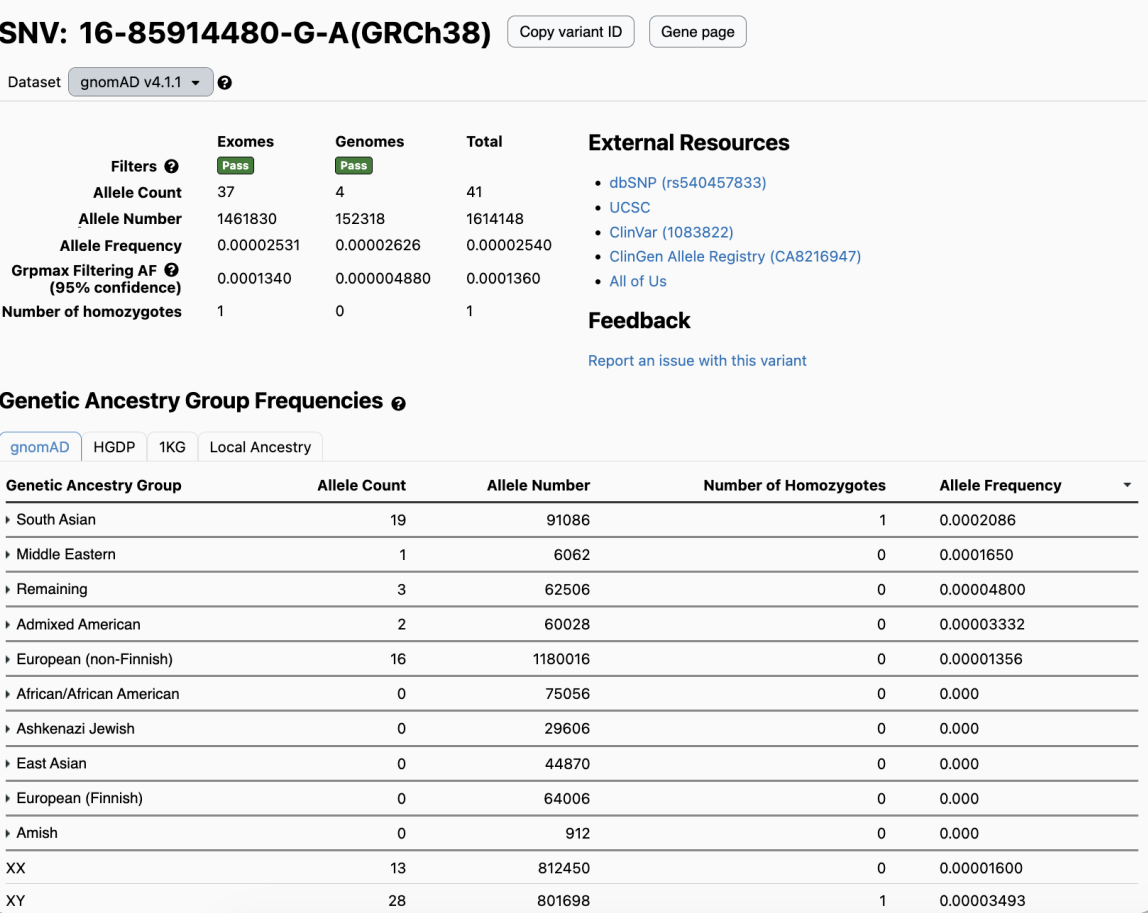
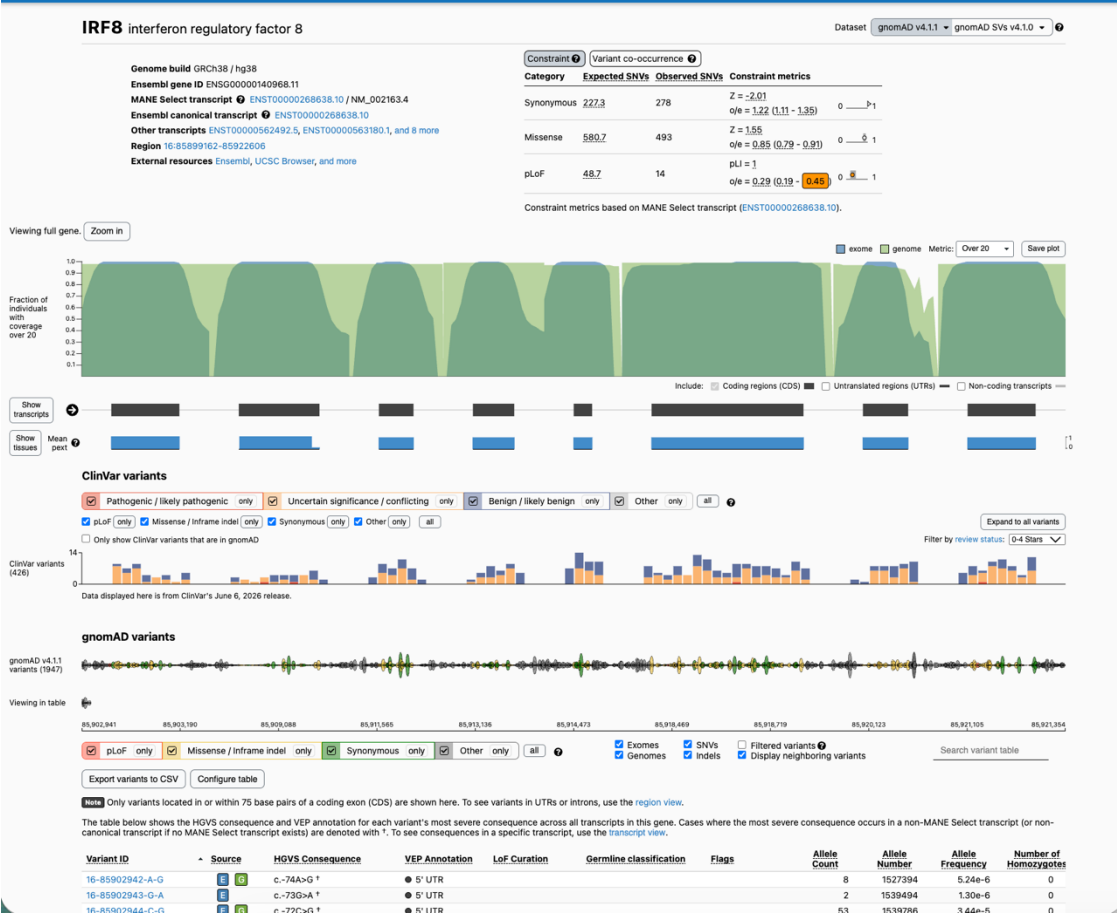
Classification [?] (Last evaluated)	Review status [?] (Assertion criteria)	Condition [?]	Submitter [?]	Collapse all rows [?]
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Publications: PubMed: 17661172, 21989022, 22762150				
Comment: The c.5191G>T (p.Glu1731*) variant of the BRCA1 gene creates an early stop codon. It is expected to result in an absent or disrupted protein product. This variant has not been identified in the general population by the Genome Aggregation Database (gnomAD). Truncating variants in BRCA1 are known to be pathogenic (PMID: 21989022, 17661172, 22762150). Therefore, the c.5191G>T (p.Glu1731*) variant of the BRCA1 gene is classified as pathogenic. (less)				
Observation 1 Collection method: clinical testing Allele origin: germline Affected status: unknown Number of individuals with the variant: 1 Zygosity: 1 Single Heterozygote				
Pathogenic (Jul 07, 2021) Contributing to	★ ★ ★ ★ ★ (Ambry Variant Classification Scheme 2023)	Hereditary cancer-predisposing syndrome	Ambry Genetics Accession: SCV000665042.5 First in ClinVar: Jan 01, 2018 Last updated: May 01, 2024	

gnomAD: GeNOMe Aggregation Database

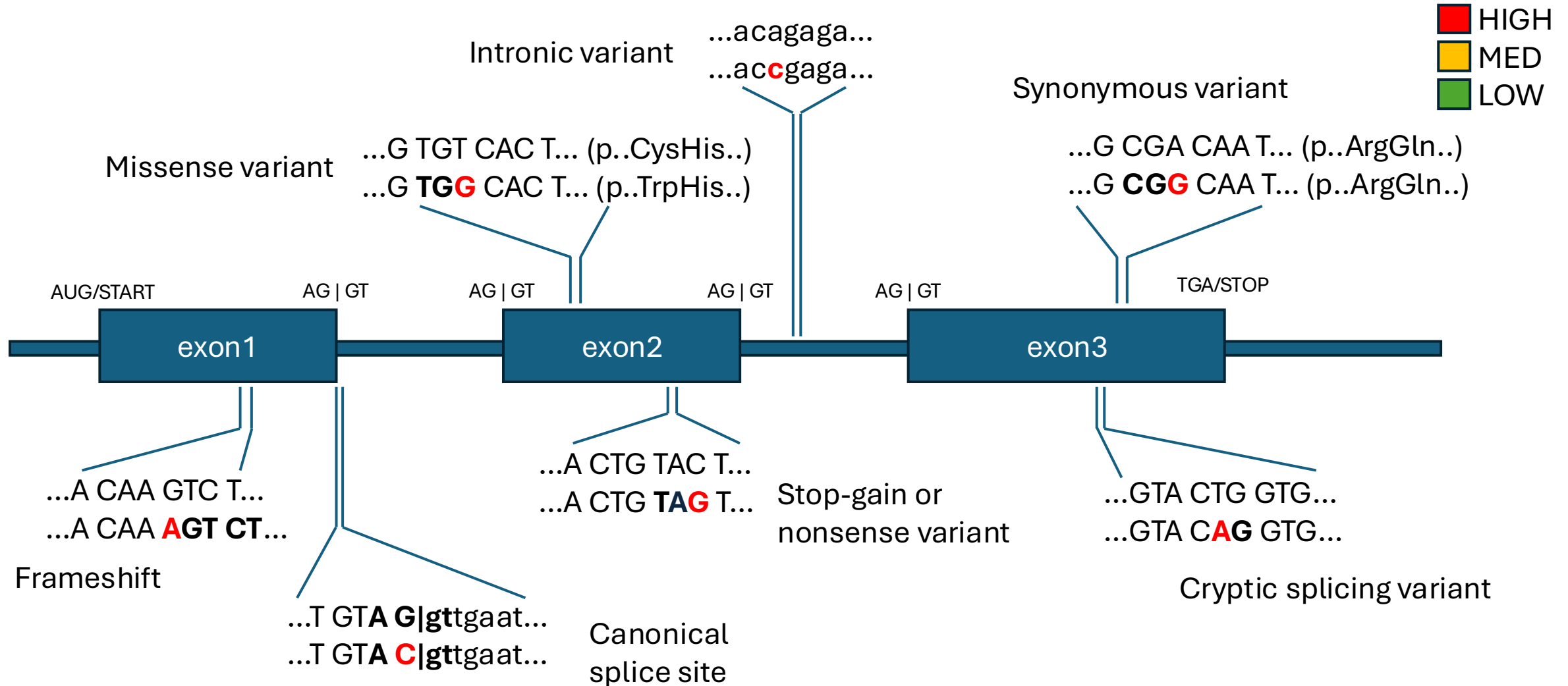
- Approximately 800,000 people, largely healthy (not known to have rare, monogenic disease)
- European ancestry enriched, but includes many thousands of individuals of African, Southeast Asian, East Asian, Admixed American (South American and Latino), and Middle Eastern populations
 - Assembled opportunistically from genomes sequenced for other genetic studies, joint processed and presented as summary statistics
- <https://gnomad.broadinstitute.org/>



gnomAD resource

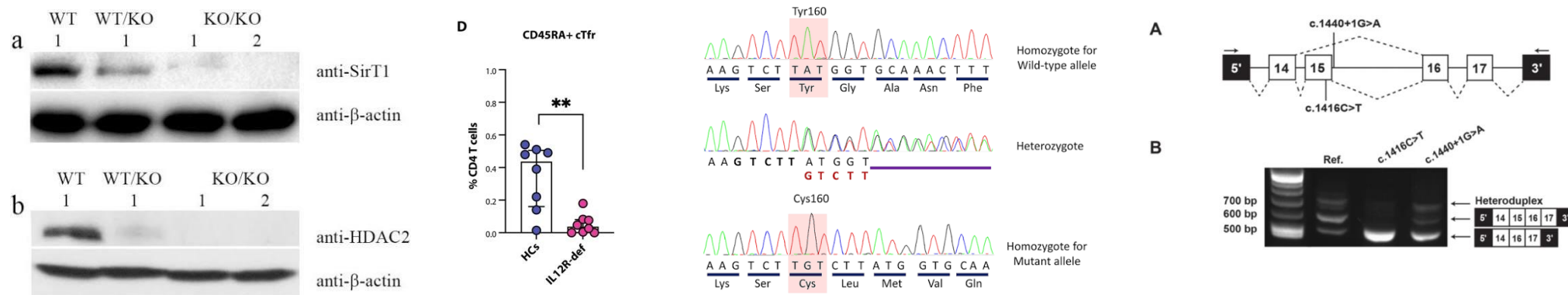


Variant annotation: how to know if a variant is functional



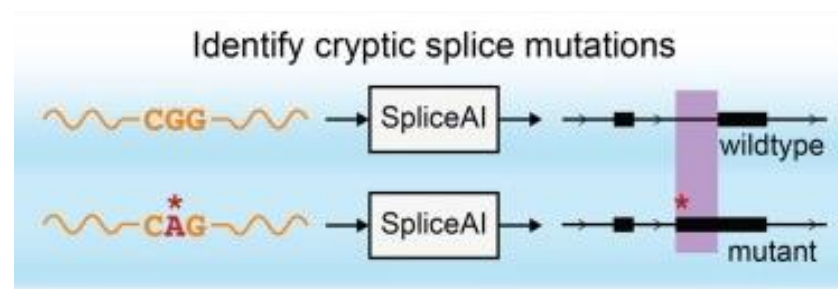
Variant annotation: how to know if a variant is functional // LOSS-OF-FUNCTION

- Predicted loss of function (pLOF) variants includes deletions, stopgain, frameshift, and canonical splice site disruption
- Variants with these consequences are expected to completely prevent the gene from making its functional gene product
- These are considered very high impact and are almost always going to be ‘qualified variants’ when you want to evaluate whether disturbed gene function is associated with any biological or clinical outcomes



Variant annotation: how to know if a variant is functional // SPLICE SITE

- SpliceAI
- Calculates the difference in predicted spliceosome affinity and recognition for the reference and variant sequences; a large difference is expected to result in the variant out-competing the reference sequence
 - A variant that out-competes the reference sequence can lead to exon skipping, exon truncation, intron retention, or pseudoexon inclusion
 - Any of these effects is considered as severe as a pLOF variant

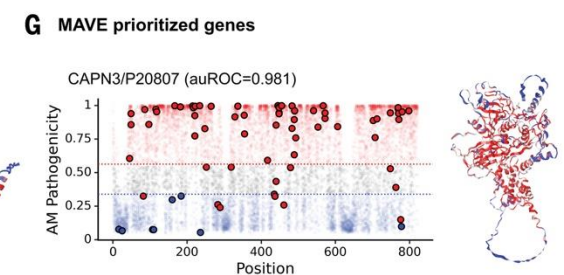
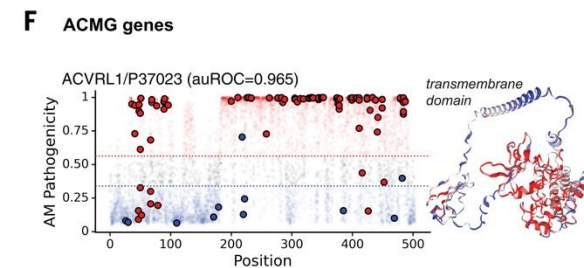
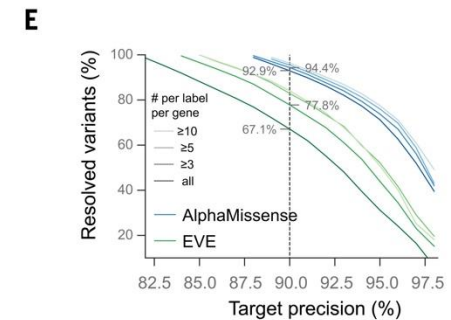
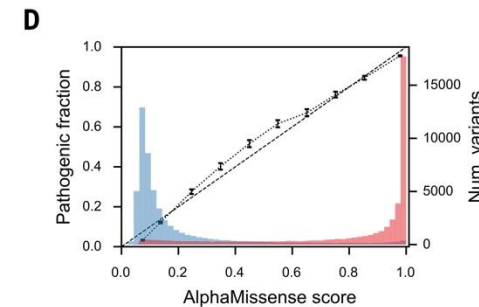


Variant annotation: how to know if a variant is functional // MISSENSE

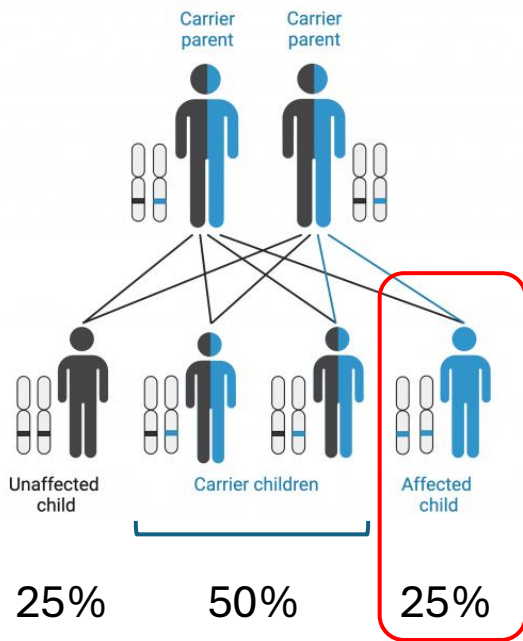
- Missense variants change one amino acid of a protein-coding gene
 - folding and structure
 - function of an active site or binding interface
 - stability of the protein
 - capacity for post-translational modifications or proper targeting
- Many missense variants are compatible with normal function and some are very common, but other missense variants will create a non-functional or unstable protein
- There are many bioinformatics algorithms designed to predict the deleteriousness or damaging-ness of a missense substitution
 - SIFT, PolyPhen, REVEL, PrimateAI, AlphaMissense, VEST4, FATHMM
- As a class, this category needs to be filtered into predicted-damaging and predicted-tolerated. The predicted damaging subset are typically going to be qualified for a burden test or other genotype-phenotype study

Variant annotation: how to know if a variant is functional // MISSENSE

- AlphaMissense
- Deep learning model that predicts the pathogenicity of missense variants by learning patterns from protein sequence, structure, and evolutionary conservation across species,
- Outputs a score that classifies variants as likely benign, likely pathogenic, or uncertain based on comparison between the variant and the reference amino acid
- Many other algorithms also exist



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Burden test workflow

1. Use domain expertise to decide on parameters and key covariates
2. Annotate all observed variants with consequence, frequency, and bioinformatics predictions
3. Apply filters to determine the qualified variant set
4. Conduct statistical analysis to determine if cases and controls are asymmetric in their qualified variant carrier burden for any genes

Note: ancestry stratification

- Controlling for relatedness and stratification within the analysis-ready cohort is ESSENTIAL for accurately conducting genotype-to-phenotype studies
- People who are more similar in general will be more similar at every gene, not just the gene(s) that are truly associated with your trait
 - This can create false positives (genomic inflation)
- Diseases may have different genetic architecture in different populations, especially if you have founder effects (genetic heterogeneity; slide 7)
- Best approach: consider ancestry and any known relatedness while recruiting cases and controls
- Next best approach: incorporate principal components as covariates
 - For a highly multi-ethnic cohort, you may need to go further and use meta-analysis techniques